PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

cerebral palsy and neuromuscular

1. Cerebral palsy: definition and classification

- A group of permanent disorders of movement and posture caused by a **non-progressive lesion of the immature brain** (before, during or within about 2 years after birth). The brain lesion is static, but the **musculoskeletal problems are progressive** (muscles do not grow at the rate of bone under spasticity, so contractures and bony deformity develop with growth).
- **Risk factors:** prematurity and low birth weight (strongest), perinatal hypoxia, intrauterine infection, multiple pregnancy, kernicterus (athetoid CP), neonatal stroke.
- **Physiological classification:**
  - **Spastic** (about 70-80%; upper motor neurone, velocity-dependent tone; pyramidal tract).
  - **Dyskinetic** (athetosis, dystonia, chorea; basal ganglia; kernicterus). **Poor candidates for tendon surgery** (unpredictable results).
  - **Ataxic** (cerebellum).
  - Mixed.
- **Topographic classification: hemiplegia** (arm worse than leg; walks by about 18-24 months; seizures common), **diplegia** (legs much worse than arms; classic for prematurity with **periventricular leukomalacia**), **quadriplegia / total body involvement** (all limbs, poor head control, intellectual disability, seizures, feeding problems).
- **GMFCS (Gross Motor Function Classification System), age 6-12 descriptors:**

Level	Function
I	Walks without limitation; difficulty only with advanced skills
II	Walks with limitations (long distances, uneven ground, stairs need a rail)
III	Walks with a hand-held mobility device (walker, crutches); wheelchair for long distances
IV	Self-mobility limited; powered mobility; may walk short distances with assistance
V	Transported in a manual wheelchair; very limited head and trunk control

- GMFCS is stable over time and predicts **hip displacement risk** (about 0% at level I to about 90% at level V) and walking prognosis.
- **Walking prognosis:** sitting independently by **age 2** predicts walking; not sitting by 4 suggests the child will not walk. Persistent primitive reflexes after age 1 (Bleck's criteria) are poor signs.

2. Principles of management

- **Team approach:** physiotherapy, orthoses, tone management, orthopaedic surgery. Goals depend on GMFCS: improve gait in I-III; keep sitting comfortable, hips painless, and spine balanced in IV-V.
- **Tone management:**
  - **Botulinum toxin A** (blocks acetylcholine release at the neuromuscular junction) for focal **dynamic** spasticity in young children (typically 2-6), effect lasts about 3-6 months. Combine with serial casting. Not useful for fixed contracture.
  - **Oral baclofen** (GABA-B agonist; sedation limits use); **intrathecal baclofen pump** for severe generalised

spasticity or dystonia (GMFCS IV-V).
    - **Selective dorsal rhizotomy (SDR):** sectioning a proportion (often about half) of the L2-S1 dorsal rootlets, guided by intraoperative stimulation. Ideal candidate: **spastic diplegia, GMFCS II-III, age about 4-8**, good strength and selective motor control, minimal fixed contractures, no dystonia. Permanent reduction of spasticity; may unmask weakness; later hip subluxation and spinal deformity can occur.
- **Gait analysis** (3D kinematics, kinetics, EMG) before multilevel surgery.
- **Single-event multilevel surgery (SEMLS):** correct all soft-tissue and bony deformities in one operation (rather than "birthday syndrome" of one operation a year), typically at **age about 6-10** when gait has matured. Followed by intensive rehabilitation for about a year.

3. The hip in CP

- **Spastic hip displacement** is due to overactive **adductors and iliopsoas** with weak abductors and extensors, leading to **coxa valga, increased femoral anteversion, and acetabular dysplasia (posterosuperior)**. The hip subluxes **posterosuperiorly** (unlike DDH).
- **Hip surveillance:** clinical exam and AP pelvis radiograph from about age 2, every 6-12 months, more often for GMFCS III-V (Australian/CPUP-based protocols).
- **Reimers migration percentage (MP):** percentage of the femoral head width lateral to Perkin's line. Normal under about 10%; **over 30-40% = subluxation**; 100% = dislocation. **Progression over about 10% a year** is significant.
- **Management by severity:**

Situation	Surgery
Young child (under about 6-8), **MP 30-40%** (some use 25-60%)	**Preventive soft-tissue release:** adductor longus and gracilis tenotomy ± iliopsoas lengthening (psoas over brim)
MP over 40-50%, older child, or failed release	**Reconstructive: varus derotation osteotomy (VDRO)** of the proximal femur ± **pelvic osteotomy (Dega or San Diego)** for acetabular dysplasia, with open reduction if needed
Painful, long-standing dislocation with arthritic head in a non-walker	**Salvage:** proximal femoral resection (Castle or McHale), valgus osteotomy (Schanz), arthrodesis or arthroplasty

- Do not do an isolated obturator neurectomy (anterior branch) routinely; abandoned by most because of over-abduction (windswept or abduction contracture).
- **Windswept hips:** one hip abducted, the other adducted and dislocated, with pelvic obliquity and scoliosis.

4. The knee in CP

- **Crouch gait** (excessive knee flexion in stance): due to hamstring contracture, weak soleus or plantarflexor-knee-extension couple insufficiency (often **after isolated over-lengthening of the Achilles tendon**), lever-arm dysfunction (external tibial torsion, foot instability). Treat with hamstring lengthening, correction of lever arms, **distal femoral extension osteotomy** with **patellar tendon advancement** in severe fixed crouch, and floor-reaction AFOs.
- **Stiff-knee gait:** reduced and delayed knee flexion in swing from an **overactive rectus femoris** (confirmed by dynamic

EMG; Duncan-Ely test). Treat with **distal rectus femoris transfer** (to the semitendinosus or gracilis) rather than release.

- **Hamstring tightness:** popliteal angle increased. Medial hamstring lengthening; lateral lengthening only if needed (risk of recurvatum and crouch). **Hamstring lengthening increases anterior pelvic tilt** (the hamstrings are hip extensors too).
- **Jump gait** (equinus + knee and hip flexion) and **true equinus** (hemiplegia). The Rodda and Graham sagittal gait classification: true equinus, jump, apparent equinus, crouch.

5. The foot in CP

- **Equinus** is the commonest deformity.
  - **Silfverskiold test:** if dorsiflexion improves with the knee flexed, the **gastrocnemius alone** is tight: **gastrocnemius recession** (Strayer, Vulpius, Baker). If not, both are tight: **tendo-Achilles lengthening** (Hoke triple hemisection, Z-lengthening).
  - **Avoid over-lengthening the Achilles tendon** in diplegia: it causes **crouch gait**, which is far worse than mild equinus. Diplegic equinus is often "apparent" (from knee and hip flexion).
- **Equinovarus** (hemiplegia): overactive **tibialis posterior** (split posterior tibial tendon transfer, SPLATT-type to peroneus brevis, or lengthening) or **tibialis anterior** (split anterior tibial tendon transfer, **SPLATT**). Confirm the culprit with dynamic EMG.
- **Equinovalgus / planovalgus** (diplegia and quadriplegia): peroneal overactivity and gastrocnemius-soleus contracture. **Calcaneal lengthening osteotomy (Evans)** in mobile feet; **subtalar or triple arthrodesis** in rigid or severe feet in older children.

- **Hallux valgus** develops with planovalgus: first MTP arthrodesis.

6. Spine and upper limb in CP

- **Neuromuscular scoliosis:** long **C-shaped thoracolumbar curves** with **pelvic obliquity**, mainly in GMFCS IV-V. Progresses even after skeletal maturity (especially curves over about 40 degrees).
- **Posterior spinal fusion from T2 to the pelvis** (pelvic fixation with iliac or **S2-alar-iliac** screws) when the curve exceeds about 40-50 degrees and sitting balance worsens. Complications are higher than in idiopathic scoliosis: infection, blood loss, pulmonary problems. Optimise nutrition first.
- **Upper limb (hemiplegia):** shoulder internal rotation, elbow flexion, forearm pronation, wrist and finger flexion, **thumb-in-palm**. Surgery (e.g. **flexor carpi ulnaris to ECRB transfer**, pronator teres rerouting, thumb adductor release) mainly improves function and hygiene; needs good sensation and voluntary control (the House and Zancolli classifications).

7. Muscular dystrophies

Duchenne muscular dystrophy (DMD)

- **X-linked recessive**, absent **dystrophin** (Xp21; deletions in about 65%); one-third are new mutations. Boys; incidence about 1 in 3,500-5,000 male births.
- Presents at **3-5 years**: clumsiness, frequent falls, **toe-walking**, **Gowers sign** (climbing up the thighs to stand: proximal weakness), **calf pseudohypertrophy** (fat and fibrous replacement), waddling gait, lumbar lordosis.
- **CK massively raised** (10-100 times normal, even at birth); diagnosis by **genetic testing** (multiplex ligation-dependent

probe amplification); muscle biopsy (absent dystrophin on immunostaining) if needed.

- **Natural history:** loss of walking around **age 12** (later on steroids), scoliosis after loss of ambulation, death in the 20s-30s from **cardiomyopathy** or respiratory failure.
- **Corticosteroids** (prednisolone or deflazacort) prolong walking by about 2-3 years, reduce scoliosis and improve cardiorespiratory function. Exon-skipping drugs (e.g. eteplirsen for exon 51) and gene therapy for selected mutations.
- **Orthopaedics:** keep walking as long as possible (Achilles and ITB releases with KAFOs in selected cases). **Scoliosis:** fuse **early**, when the curve is **about 20-30 degrees**, before the **FVC falls below about 30-35%** (anaesthetic risk rises sharply). Long fusion to the pelvis or L5. **Malignant hyperthermia-like reactions** and rhabdomyolysis: avoid succinylcholine and volatile agents.
- Fractures from osteoporosis (steroids, immobility): treat quickly to keep walking.

Becker and other dystrophies

- **Becker:** reduced or abnormal dystrophin; later onset (over about 7), walks beyond 16, milder, cardiomyopathy common.
- **Limb-girdle** (autosomal), **facioscapulohumeral** (AD; scapular winging, facial weakness: **scapulothoracic fusion** for winging), **Emery-Dreifuss** (early **elbow flexion contractures**, Achilles contracture, **cardiac conduction block**: pacemaker).
- **Myotonic dystrophy** (CTG repeat; distal weakness, myotonia, cataracts, frontal balding, cardiac conduction defects).

8. Spinal muscular atrophy (SMA)

- Autosomal recessive **SMN1 deletion**, loss of **anterior horn cells**: symmetric proximal weakness, absent reflexes, tongue fasciculation, **normal sensation and intelligence**. SMN2 copy number modifies severity.
- **Types:** I (Werdnig-Hoffmann; never sits, death in infancy without treatment), II (sits but never stands), III (Kugelberg-Welander; walks), IV adult.
- **Nusinersen** (antisense oligonucleotide, intrathecal), **onasemnogene abeparvovec** (gene therapy) and **risdiplam** (oral) have transformed survival.
- **Scoliosis** almost universal in type II: bracing for sitting, then **growth-friendly instrumentation** (growing rods, magnetically controlled) or fusion; leave access for intrathecal therapy if possible.
- **Hip subluxation and dislocation** are common but usually **painless: leave them alone**.

9. Myelomeningocele (spina bifida)

- Folic acid deficiency, valproate. Associated **Arnold-Chiari II malformation, hydrocephalus (shunt)**, tethered cord, syringomyelia, neurogenic bladder, **latex allergy** (latex-free theatre).
- **Motor level determines function:**

Level	Key muscle working	Typical function
Thoracic / L1-L2	Hip flexors only	Wheelchair; standing in a frame

Level	Key muscle working	Typical function
L3-L4	**Quadriceps (L3-4), tibialis anterior (L4)**	Household or community walker with AFOs/crutches; **functional quadriceps is the key to walking**
L5	Hamstrings, toe extensors, gluteus medius partly	Community walker, AFOs
Sacral	Gastrocnemius-soleus	Near-normal walking

- **Hip dislocation** is commonest at L3-L4 (unopposed hip flexors and adductors vs weak abductors and extensors). **Do not reduce** a dislocated hip in most cases; correct contractures to keep the pelvis level.
- **Rapidly increasing scoliosis, new spasticity, deteriorating gait, changing foot deformity or urinary function:** suspect **tethered cord, shunt malfunction or syrinx**; MRI and neurosurgical review.
- **Feet:** aim for a **plantigrade, braceable foot**. Rigid clubfoot (talectomy in severe cases), vertical talus, calcaneus foot (L4-L5; anterior tibial transfer to the calcaneus). **Insensate feet** develop pressure ulcers and Charcot changes.
- **Fractures** in insensate limbs present as warm, swollen limbs with fever (mistaken for infection) and heal with exuberant callus.

10. Hereditary neuropathies and other conditions

- **Charcot-Marie-Tooth (HMSN):** commonest inherited neuropathy. **Type 1A: PMP22 duplication on chromosome 17**, demyelinating, slow conduction velocities. Type 2: axonal.
  - **Cavovarus foot**, claw toes, peroneal muscle wasting ("**inverted champagne bottle**" legs), intrinsic hand

weakness.
    - **Mechanism:** weak **tibialis anterior** is overpowered by **peroneus longus** (plantarflexed first ray, forefoot-driven cavus); weak **peroneus brevis** is overpowered by **tibialis posterior** (hindfoot varus).
    - **Coleman block test:** a block under the lateral forefoot lets the first ray drop; if the **hindfoot varus corrects**, the hindfoot is flexible and the deformity is forefoot driven: plantar fascia release, **first metatarsal dorsiflexion osteotomy**, **peroneus longus to brevis transfer**, ± tibialis posterior transfer. If not, add a **lateralising calcaneal osteotomy**; triple arthrodesis only for rigid, arthritic feet.
- **Friedreich ataxia:** AR, **GAA trinucleotide repeat in frataxin** (chromosome 9). Ataxia from about age 10, areflexia, cavovarus feet, scoliosis (more like idiopathic in shape), **hypertrophic cardiomyopathy** (the main cause of death), diabetes.
- **Arthrogryposis multiplex congenita:** multiple **non-progressive** congenital contractures, normal sensation and usually normal intelligence. **Amyoplasia** (classic): internally rotated shoulders, extended elbows, flexed wrists, rigid **clubfeet**, dislocated hips, knee contractures. Early stretching; surgery aims for walking and independent function (bilateral hip dislocations are usually left; unilateral may be reduced).
- **Poliomyelitis (residual paralysis)** is still examined in India: asymmetric flaccid paralysis with normal sensation; tendon transfers once power has plateaued (about 2 years), "**a muscle loses one grade of power after transfer**", and transfer **in phase** where possible.

Exam pearl: **Silfverskiold test positive = gastrocnemius recession; over-lengthening the Achilles in a diplegic child causes crouch gait. Stiff-knee gait = distal rectus femoris transfer.**

Exam pearl: **DMD scoliosis: fuse early (curve about 20-30 degrees, FVC above about 30-35%). SMA hip dislocation: leave alone. Myelomeningocele: L3-L4 (quadriceps) is the key level for walking.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.